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الباقيات
المصالحات

Learning objectives

- why Dense network doesn't suitable for pictures or images
- what is CNN
- convolution
- Filter / Kernel
- Feature Map
- Stride
- Padding
- parameter sharing
- Feature hierarchy

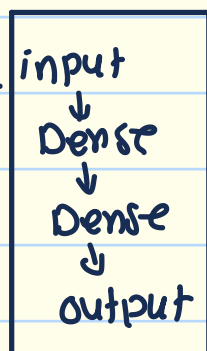
why Dense network doesn't suitable for pictures or images

الفكرة ببساطة احنا كبشر بنشوف صورة الجلد طبيعي وعالي لكن الكمبيوتر مستحيل يستوعبها كم صورة
احنا يستوعبها كأرقام ولو عنا صورة : $224 \times 224 \times 3$ ولا غل RGB : الصورة الواحدة

رقم 150,528

$\therefore 224 \times 224 \times 3$

طيب المشكلة ان Dense Network كذا NN



* المشكلة الأولى

ان كل input مربوط مع كل

neuron في ال next layer

ولو فرضنا ان input 150,528

وحظنا في اول Dense layer

120 neuron فبنا 20 مليون

weight حسب في اول Dense

layer

* المشكلة الثانية

المشكلة مش بس بار weight في الصور

الاحتمالية زي صورة الجلد ان pixels القريبة من بعض

مكان تكون Texture , Line , Edge لكن لو عملنا Flatten الصورة ماسقة

بتحول تقريباً ل array من الأرقام وار Dense بتعاطل معها كقائمة من الأرقام

ففي بعض ما عندها فكرة واضحة انق هار ان pixel موجود بجانب pixel

نار Dense بتجاهل ال spatial structure فهون احب فكرة ال CNN

ان الصورة مش مجرد List ارقام

what is CNN

CNN · Convolutional Neural Network

فكرتها مختلفة بل ما نقول اعطيني كذا ال 150 ألف رقم دفعة وحدة بتقول خليني اطلع كل

احداه مخيدة من الصورة مثلاً pixel 3x3 وتستخدم شئ اسمه Filter / Kernel

زي مثلاً : - 1 0 1

- 1 0 1

- 1 0 1

← كارت Grid
كش مربعة
مربعة

ها 3x3 Kernel يعني فيها 9 numbers

وظيفة انه يبحث عن Pattern معين وبعدها يرد

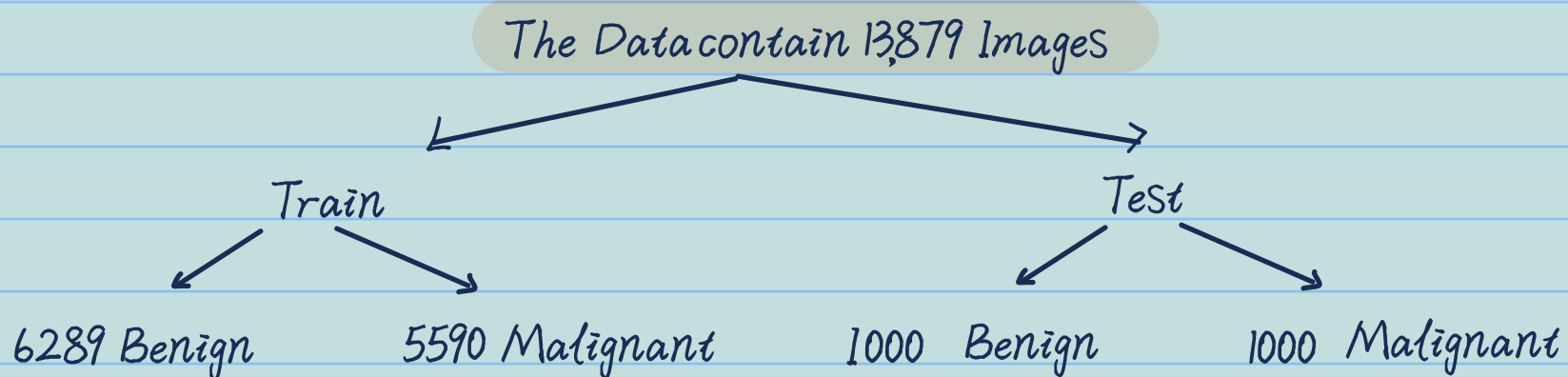
اجيب 3x3 وهكذا

Filter / Kernel , the main Goal explore the vertical edges

our project.

Input: skin lesion Images

output:- Benign or Malignant



. Binary Image classification, because we have two classes

so Dense networks are inefficient for images, because an image contains a very large number of pixel values, which creates millions of weights

They also do not naturally exploit the spatial relationships between neighboring pixels

Kernel/filter : A small grid / Matrix of weights that detect one pattern

For Example:-

$$\begin{bmatrix} \overset{\text{Left}}{-1} & 0 & 1 \\ -1 & \overset{\text{Middle}}{0} & 1 \\ -1 & 0 & \overset{\text{right}}{1} \end{bmatrix}$$

It is 3x3, in our project use it as a vertical Edge Detector

How it works: focusing on a very small area, not whole picture Like:- 3x3 pixel
And by asking: Is the pattern I'm looking for there

brightness on left
—
brightness on right

if the left & right are close
then response ≈ 0 → means there is no changes

if the left & right ones dark and another Bright
then response become large → means there is edges

Convolution :

Suppose we have a small Area from image and Kernel

$$\begin{array}{c} \text{Small Area} \\ \begin{bmatrix} 1 & 2 & 5 \\ 1 & 2 & 5 \\ 1 & 2 & 5 \end{bmatrix} \end{array} \quad \begin{array}{c} \text{Kernel} \\ \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix} \end{array}$$

$$\begin{array}{l} 1 \times (-1) + 2 \times 0 + 5 \times 1 = 4 \\ 1 \times (-1) + 2 \times 0 + 5 \times 1 = 4 \\ 1 \times (-1) + 2 \times 0 + 5 \times 1 = 4 \end{array} \quad \left. \vphantom{\begin{array}{l} 1 \times (-1) + 2 \times 0 + 5 \times 1 = 4 \\ 1 \times (-1) + 2 \times 0 + 5 \times 1 = 4 \\ 1 \times (-1) + 2 \times 0 + 5 \times 1 = 4 \end{array}} \right\} \therefore 12 \quad \begin{array}{l} \text{means that the Kernel clearly} \\ \text{responded to this area} \end{array}$$

if the whole area similar

$$\begin{array}{c} \text{Small Area} \\ \begin{bmatrix} 5 & 5 & 5 \\ 5 & 5 & 5 \\ 5 & 5 & 5 \end{bmatrix} \end{array} \quad \begin{array}{c} \text{Kernel} \\ \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix} \end{array}$$

$$\begin{array}{l} 5 \times (-1) + 5 \times 0 + 5 \times 1 = 0 \\ 5 \times (-1) + 5 \times 0 + 5 \times 1 = 0 \\ 5 \times (-1) + 5 \times 0 + 5 \times 1 = 0 \end{array} \quad \left. \vphantom{\begin{array}{l} 5 \times (-1) + 5 \times 0 + 5 \times 1 = 0 \\ 5 \times (-1) + 5 \times 0 + 5 \times 1 = 0 \\ 5 \times (-1) + 5 \times 0 + 5 \times 1 = 0 \end{array}} \right\} \therefore 0 \quad \begin{array}{l} \text{There is no important difference between Left \&} \\ \text{Right, there is no clear vertical Edges} \end{array}$$

So, that why the kernel called feature Detector
Because it looking for specific feature

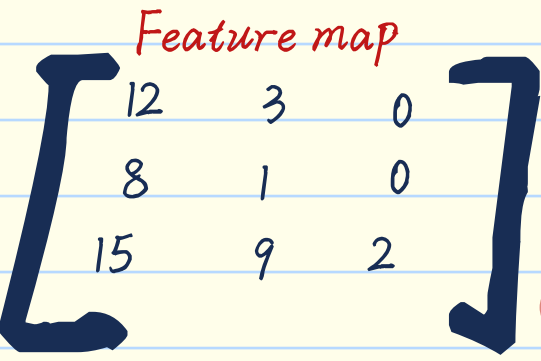
The kernel doesn't stop here

calculate first area then move then calculate second area then more until cover whole image

calculation + moves called convolution

Feature Map

each position gives one number ← فنجد ما نمشي على الصورة كلها بتجميع الأرقام في matrix



Feature map tell us where the pattern appear in Image

- ** High value → Filter responded aggressively
- ** Low value → weak or non-existent feature

In our project: feature map (222x222)

Input = 224
Kernel : 3
Stride = 1
Padding = valid

output = Input - kernel + 1
224 - 3 + 1 = 222

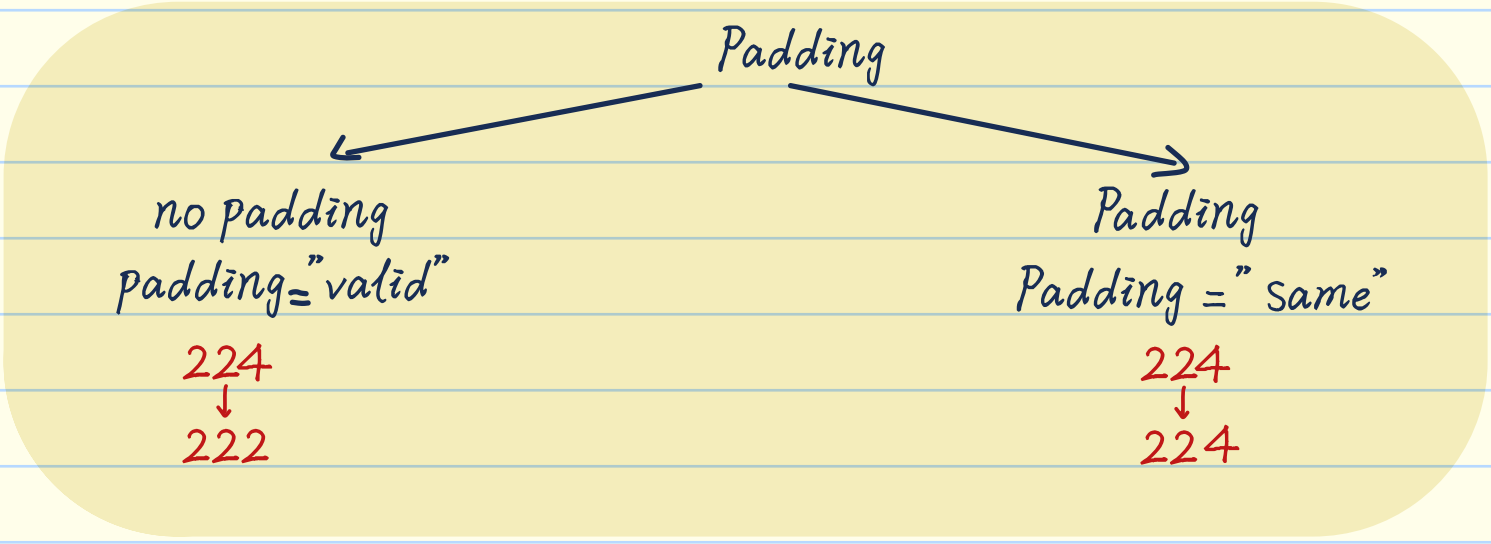
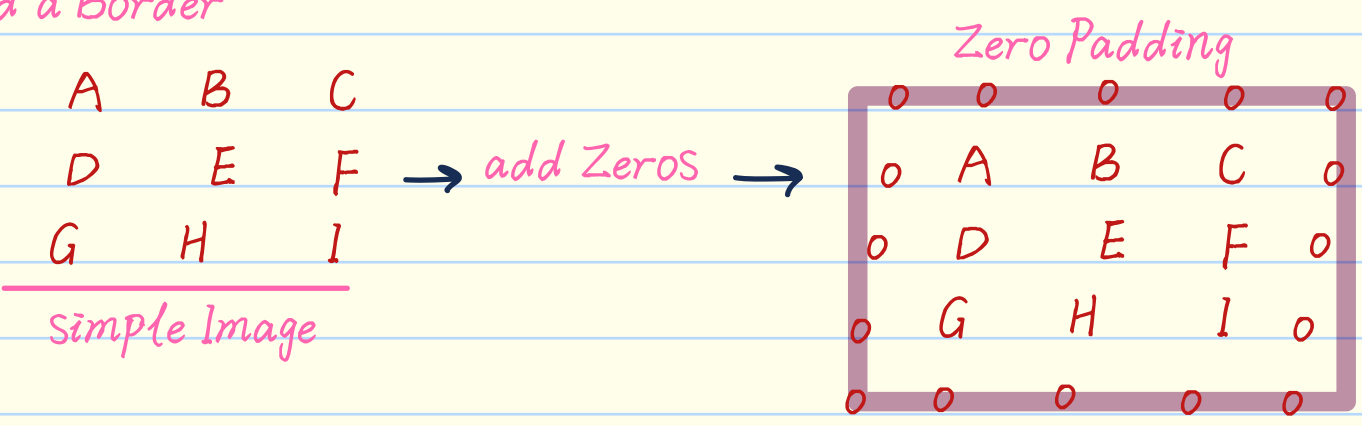
نقوسنا كذا ال kernel
من المساحة ليطلع بها
الحدود

Padding :-

if each convolution layer reduces the image 224 → 222 → 220 → 218 → ...
we can see spatial information quickly

The solution · add a Border

For example



Stride $\rightarrow$ How many steps dose the filter moves each time

if Stride = 1 $\rightarrow$ single pixel 1 $\rightarrow$ 2 $\rightarrow$ 3 $\rightarrow$ ---

if Stride = 2 $\rightarrow$ twice pixel 1 $\rightarrow$ 3 $\rightarrow$ 5 $\rightarrow$ ---

Parameter Sharing

means the same parameters are reused throughout the image

This is one of the main reason, that can needs much less parameters than Dense

for example:-

Kernel

$$\begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}$$

then 9 values, these value are

used in top-left then center

then bottom-right

* Don't make a new kernel for each position

In Day 1 we use

$$\begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}$$

to understand the Idea

but in CNN

Filter start with wight



Training



Back propagation



wights change



filter learns usful patterns

So the network may be learn filter that can responsed to: edger, textures, curves

one filter dosen't enough

For example: if we have $\text{conv2D}(32, (3,3)) \rightarrow$ means 32 filters



each filter produce 1 feature map

$\therefore$ 32 feature map

Filter learn 32 pattern and feature map display where the feature exist and how many pattern

Feature hierarchy

